# ADINA NADEEM

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## SUMMARY

MSc Bioinformatics student at Freie Universität Berlin with hands-on experience in statistical data analysis, reproducible pipelines, and scientific programming in Python and R. Skilled in data cleaning, exploratory analysis, and visualization, with a strong foundation in genomics and statistical analysis, coupled with a knack for learning new tools and collaborating across interdisciplinary teams. Passionate about applying computational methods and AI-driven approaches to advance biomedical research and deliver actionable insights. Eager to contribute to projects that integrate clinical, genomic, and patient-reported data to drive innovations in precision medicine.

## EDUCATION

**Freie Universität Berlin** **Oct 2024 - Present**
Master of Science in Bioinformatics – Specialization in Data Science
**COMSATS University, Islamabad**
Bachelor of Science in Bioinformatics

## TECHNICAL SKILLS

- **Languages:** Python, R, Bash, C++, SQL (Postgres), JavaScript, HTML/CSS, Rust (beginner)
- **DevOps & Infrastructure:** Docker, AWS (EC2, S3), Git/GitHub/GitLab, CI/CD concepts
- **Bioinformatics Tools:** Nextflow, Trimmomatic/fastp, Salmon, SAMtools, DESeq2, Seurat, Biopython, BowTie2, BWA, BLAST, gatk, VEP, Picard, Chimera, GROMACS
- **Operating Systems:** Linux (PopOS, Mint, Ubuntu), Windows
- **AI & Data:** Basic experience with machine learning workflows, statistical analysis, and exploratory data visualization; interest in LLM applications

## RESEARCH EXPERIENCE

**Development of Workflow for the Detection of Polyadenylated Reads from RNA-Seq Data** **Aug 2022 – Feb 2023**
***Research Assistant, COMSATS University, Department of Biosciences***
- Designed and implemented a highly efficient Nextflow-based pipeline, utilizing Bash scripting and Python to effectively filter out mRNA transcripts (polyadenylated reads) from raw data.
- Analyzed gigabytes of RNA sequencing data using Python and created visually compelling graphs to demonstrate data quality.
- Created a Docker image to ensure platform independence and maximize compatibility with various operating systems.

## ACADEMIC PROJECTS

***"Bacterial Genome Assembly & Annotation" - Ongoing***
- Developed and optimized a Snakemake workflow to assemble, polish and annotate bacterial genomes from paired-end Illumina and long-read ONT data.
- Implemented per-sample logic for hybrid and short-read assembly using SPAdes/Unicycler, performed stringent QC with fastp/NanoFilt, and executed polishing steps (Racon → Medaka → Pilon).
- Added downstream modules for gene annotation (Bakta), pan-/core-genome inference (Panaroo) and phylogenetic analysis (IQ-TREE 2), with optional screens for MLST, AMR genes, virulence factors and plasmids.
- Configured resources for HPC execution and ensured reproducibility through conda environments and YAML configuration.

***"Metagenome Assembly & Analysis Workflow" [GitHub]***
- Designed and implemented an end-to-end Snakemake pipeline for complex metagenomic datasets, performing QC, assembly (megahit), polishing (racon), and taxonomic classification (kraken2).
- Developed a novel sub-workflow to automatically select the best reference genome using Mash and RagTag for comparative analysis.
- Demonstrated ability to orchestrate multi-step genomic analyses and summarize results effectively using MultiQC.

***"Transcript Quantification Workflow" [GitHub]***
- Built a flexible RNA-seq pipeline using Snakemake to perform transcript quantification with a choice of alignment-based (STAR, RSEM, featureCounts) and alignment-free (Kallisto) methods.
- Engineered a modular design controlled by a simple sample sheet, allowing for easy configuration of tools, parameters, and inputs.
- Showcased experience in integrating outputs from diverse bioinformatics tools into a unified and comprehensive report.

## ACHIEVEMENTS & CERTIFICATIONS

- **Google Summer of Code 2024**: Proposal recommended by a Wellcome Sanger Institute researcher during the mentorship phase for strong technical merit and open-source bioinformatics expertise, despite the project not passing the final committee vote.
- **Major League Hacking (MLH) Fellowship 2024**: Advanced to the final interview stage from a competitive global applicant pool.
- **Certifications:** AWS Certified Cloud Practitioner, Google Advanced Data Analytics Professional Certificate , Google Business Intelligence Certificate, McKinsey Forward.

## SOFT SKILLS

- Leadership, Communication, Teamwork, Problem-Solving, Time Management